MASTER

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Exploring the Role of Perceived Source (Human vs. AI) on Feedback Effectiveness and Trust

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Perceptions of Feedback on Written Reflections:

Exploring the Role of Perceived Source (Human vs. AI) on Feedback Effectiveness and Trust

by Petra Slokker

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Abstract

Reflection, particularly through reflective writing, is an important part of learning, helping students understand experiences and set goals for improvement. High-quality feedback supports this process but is often time-consuming to deliver. Artificial intelligence (AI) offers a promising solution by generating personalized feedback. However, students may still perceive AI feedback as less effective than human feedback, potentially due to lower trust in AI sources. This study investigated how the perceived feedback provider (human vs. AI) affects students' perceptions of feedback effectiveness in the context of reflective writing, with a key focus on the role of trust in the feedback provider. Using a within-subject design, participants received identical feedback labeled first as human generated and then as AI-generated. Results showed that feedback believed to be generated by a human was perceived as more effective than AI-generated feedback. This effect was mediated by differences in trust in the feedback provider. Furthermore, the findings showed that familiarity with AI chatbots or frequency of use did not significantly influence trust. These results highlight the importance of trust in shaping perceptions of feedback effectiveness. This study's findings suggest that for effectively integrating AI in education, it is necessary to not only provide high-quality feedback content, but also to build and maintain students' trust in AI systems as feedback providers. This study provides valuable insights for educators and AI developers aiming to integrate AI feedback effectively into reflective learning environments.

1 Introduction

Reflection, particularly through reflective writing, has become an essential component in education. It is seen as a necessary part of learning, helping students to better understand their experiences, recognize their strengths and weaknesses, and set goals for improvement (Helyer, 2015; Taylor, 2023; Yancey, 1998). High-quality feedback can support students' reflective thinking and learning development (Moussa-Inaty, 2015). This impact is strongest when students perceive feedback on their written reflections as effective (Gamlem & Smith, 2013; Lee, 2018). However, providing effective feedback is often time-consuming and requires a lot of energy from teachers (Gan et al., 2021; Moussa-Inaty, 2015).

Artificial intelligence (AI) offers a potential solution by generating equally detailed and consistent feedback as that of human assessors (Escalante et al., 2023; Jayawardena et al., 2025). However, students may still perceive human feedback as clearer and more accurate (Şık and Pektaş, 2024) and higher in quality than AI-generated feedback (Steiss et al., 2024).

However, some aspects of previous research need to be explored further. Most studies only focus on one specific aspect of feedback perception, such as accuracy and quality of the feedback. They often do not consider perceptions on feedback effectiveness, and how multiple aspects might be affected in different ways. In contrast, the current study examines multiple dimensions of feedback effectiveness. This provides a more detailed understanding in which aspects of feedback effectiveness generative AI might need improvement in order to be found as effective as human feedback.

Furthermore, while previous studies have attempted to control for the content of feedback, it remains unclear if the reasons behind the differences in students' perceptions are because of actual differences in feedback quality between the feedback sources. The question remains whether students' beliefs about the feedback provider might have influenced these differences.

Although earlier studies have measured trust in AI and human feedback providers, they did not investigate whether this factor could be an explanation for differences in students' feedback perception. Besides, they often measured trust using only a single item. This limited the understanding of how different dimensions of trust may vary between the sources. The current study will take multiple dimensions of trust into account to increase the understanding of which aspects of trust might be most influential in shaping students' feedback perceptions.

Researching this gap is important because if aspects of trust play a role in explaining differences in feedback perceptions, then effectively integrating AI into education should not only focus on improving the feedback quality, but also about building trust in the technology providing the feedback.

The aim of this study is to examine the relationship between the perceived feedback provider (human vs. AI) and students' perceptions of feedback effectiveness in the context of reflective writing. In addition, the study investigates the role of trust in the feedback provider and its potential effect on feedback effectiveness perceptions. Insights from this study can help determine which factors are important for implementing AI-generated feedback in reflective writing to effectively support students learning outcomes.

1.1. Reflection

Reflection is more than just describing experience. It is a metacognitive process aimed at creating meaningful change (Efklides, 2009; Nguyen et al., 2014; Silver et al., 2023). It is an active process that requires individuals to evaluate their own learning experiences and progress, making it a personal and subjective approach (Mohamed et al., 2022; Ruijten-Dodoiu et al., 2024a). Effective reflection offers a clear structure to connect theory with practice, making learning more meaningful and practical (Gathu, 2022; Heyler, 2015). By encouraging continuous thinking and improvement, it turns passive learning into active engagement (Heyler, 2015). Additionally, using reflective and process-based learning frameworks have been shown to help students grow into innovative thinkers and problem solvers, better preparing them for future challenges (Ruijten-Dodoiu et al., 2024b).

A powerful role in the process of reflection in education is reflective writing. Reflective writing involves the structured process in which students critically evaluate their learning goals by writing about their own experiences, thoughts, and beliefs from their personal perspective (University of Southern California, 2025). It helps students engage more deeply with course material, think critically about their progress, and develop meaningful insights (Pavlovich, 2007; Sudirman et al., 2024). Additionally, reflective writing has been found to support the development of skills such as self-awareness, problem-solving, and decision-making (Sudirman et al., 2024). This is a skill that can be applied throughout life, offering lasting benefits within reasonable time requirements (Boutet et al., 2017). Reflecting on academic experiences has been shown to help students to deepen their understanding and develop skills that are important both inside and outside the classroom (Ezezika & Johnston, 2023; Procee, 2006).

Given the importance of reflective writing, it is essential to support students in developing their reflective skills effectively. Research suggests that students benefit from clear guidance during reflection, such as through specific questions or structured frameworks (Moussa-Inaty, 2015; Ruijten-Dodoiu et al., 2024b).

One model that has proven particularly helpful for reflective writing is Gibbs' Reflective Cycle, originally developed by Graham Gibbs (1988, as cited in Skarwinahyu et al., 2019). The model consists of six steps, from describing an experience to planning future actions, that guide students through thoughtful reflection. The model is commonly implemented in reflective writing practices across educational settings (Ezezika & Johnston, 2023; Nurlatifah et al., 2023; Sekarwinahyu et al., 2019), and is often favored over other models because of its clear structure (Adeani et al., 2020). Besides that, the model encourages critical thinking about one's own learning (Markkanen et al., 2020) and it is seen as a useful tool for teachers by providing a consistent framework for assessing reflections and giving feedback (Hashim et al., 2023; Nurlatifah et al., 2023).

While structured models like Gibbs' Cycle can support both students and teachers in writing and evaluating reflections, the assessment of reflective writing presents significant challenges. Some critics argue that assessing reflective writing discourages honest and open reflection (Hargreaves, 2004; Ixer, 2016). According to Hargreaves (2004), students may write what they believe their teachers want to hear instead of expressing their true thoughts because they feel pressure to perform academically. Similarly, Ixer (2016) suggests that educators run the risk of diminishing reflection's true value and effectiveness by turning a personal process by into a performative one.

However, others argue that thoughtful assessment can support the development of reflective skills (Fisher, 2003; Gathu, 2022; Ixer, 2016; Moussa-Inaty, 2015; YuekMing & Abd Manaf, 2014). According to Fisher (2003), students can gain clearer understanding of the aspects of critical reflection by providing them clear criteria and structured guidance. Shaw et al. (2018) further emphasize that assessment can give value to reflection in the eyes of students, make learning more visible, and identify areas where additional support may be needed. Moreover, since students often focus their efforts on what is assessed, their learning tends to be shaped and motivated by assessment itself (Watkins et al., 2005). In addition, research indicates that when feedback on reflection is handled well, it can have a significant impact on student performance and learning outcomes (Moussa-Inaty, 2015). Rather than focusing on summative evaluation, educators can offer constructive, formative feedback that supports students' growth without compromising the integrity of reflection as a personal and developmental process (Gathu, 2022). Supporting this, Shute (2008) argued that feedback is seen as one of the most effective ways for making assessment formative.

These differing viewpoints suggest that while formal assessment may undermine the potential benefits of reflective writing, it can also be a valuable tool for encouraging deeper learning, especially when combined with formative feedback. Since reflection is a process-oriented activity, feedback, rather than grading, may be the most powerful tool for supporting students’ reflective growth.

1.2. Feedback

Feedback serves as one of the most influential factors in student learning (Hattie & Timperley, 2007; Montgomery & Baker, 2007), influencing students' cognitive, emotional, and social engagement with their work (Langer, 2011). The effectiveness of feedback depends on when and how it is given (Hattie and Timperley, 2007). Feedback is more effective when it is provided close in time to when the writing was completed (Steiss et al., 2024). Furthermore, it should be constructive and personalized as it fosters confidence in students’ ability to improve and make progress (Gan et al., 2021; Ruijten-Dodoiu et al., 2025). This is especially important in reflective practices, where the aim is not just to evaluate past performance but also to encourage iterative learning and personal growth (Ruijten-Dodoiu et al., 2025). According to Hattie and Timperley (2007), for feedback to be effective it must be formative, which means it should be task-focused, process-oriented, or aimed at self-regulation to enhance learning. In contrast, personal praise is often ineffective.

How students interpret and respond to feedback is important for ensuring that formative assessment effectively supports learning (Gamlem & Smith, 2013). For example, Lee (2018) found that students value and trust teacher feedback far more than peer feedback, considering it more formative and hence most effective. Students perceive feedback as effective when it motivates them to take action and improve their performance (Winstone et al., 2017), elicits positive emotions (Dawson et al., 2019; Evans, 2015), contributes to their development as learners (Carless & Boud, 2018), and is adequate so it meets their learning needs (Dawson et al., 2019; Ferguson, 2011). According to Strijbos et al. (2010), the adequacy of feedback is determined by its usefulness, fairness, and acceptance.

This all makes managing feedback one of the most challenging aspects of teaching (Gan et al., 2021). Providing personalized and detailed formative feedback is very time-consuming and requires much energy from teachers (Gan et al., 2021; Moussa-Inaty, 2015; Steiss et al., 2024). Besides, there is often a shortage of teachers relative to the large student population in schools or universities, resulting in a low instructor-to-student ratio (Boud & Molloy, 2023). This in turn limits the instructors ability to provide high-quality feedback, especially in large classes (Paris, 2022; Pavlovich, 2007).

1.3. Integration of AI in education

A promising way to reduce instructor workload and provide diverse forms of feedback is through the use of generative AI (Wongvorachan et al., 2022; Lee & Moore, 2024; Harry, 2023). AI is becoming more integrated into educational contexts (Nguyen, 2023; Tahiru, 2021), partly because it can provide feedback much faster than humans (Jayawardena et al., 2025). Studies have found that AI-generated feedback, besides being timely, can improve academic performance by helping students better identify and correct their mistakes (Hayat et al., 2024; Stöhr et al., 2024). Furthermore, Hayat et al. (2024) found that personalized and timely feedback from AI can foster a growth mindset, which supports students' academic outcomes. Additionally, AI-generated feedback has been shown to create a stress-free environment that encourages help-seeking, reflection, and more engagement in the learning process (Lee & Moore, 2024). Notably, several studies reported that AI feedback can produce equally detailed and consistent feedback as, or even superior to, that of human assessors (Escalante et al., 2023; Jayawardena et al., 2025). This all suggests that generative AI can be successfully integrated into educational settings without having to compromise learning outcomes.

However, the integration of AI-generated feedback into education is not without concerns. Researchers have noted ongoing issues related to accuracy, ethical use, and data privacy in AI systems (Afroogh et al., 2024; Harry, 2023; Nguyen, 2023; Tahiru, 2021). One important limitation is that the quality of feedback often depends on how prompts are formulated. Poorly designed prompts can lead to vague, incorrect, or unhelpful feedback (Escalante et al., 2023; Strobelt et al., 2022). This means that instructors or students need to carefully think how to phrase their input, which can also be time-consuming and technically demanding. Another key concern is the "black-box" character of large language models (LLMs), which makes it difficult to understand how feedback is generated (Manohara et al., 2024). This can reduce the purpose of the feedback, especially when important criteria is missing or unclear. It can be challenging to determine whether the feedback aligns with learning goals or teacher expectations (Maity & Deroy, 2024). These concerns highlight, that it is especially important to provide clear prompts that include the instructor's own criteria, so that, even if the decision process of LLMs cannot be directly verified, the model is more likely to generate feedback aligned with the intended learning goals.

Beyond these technical concerns and limitations, students may also be hesitant to accept grades or feedback generated by an AI system, preferring to have human input and evaluation (Harry, 2023). For example, Steiss et al. (2024) found that students often perceived feedback from teachers as more accurate and higher in quality than feedback from AI. Similarly, Hein et al. (2024) reported that AI-generated feedback was rated as less accurate than human feedback and resulted in a reduced intention to seek further feedback. Supporting these findings, Şık and Pektaş (2024) found that students perceived human feedback as clearer, more accurate, and more supportive than feedback provided by AI. This suggests that, despite prior research indicating AI-generated to be consistent and detailed, students may still perceive it as less valuable than human feedback. This highlights the need to investigate which aspects of feedback are perceived as less valuable by students when it comes from generative AI compared to when it comes from a human. The current study addresses this by examining multiple dimensions of feedback effectiveness, such as affective (emotional) responses and willingness to improve. Understanding the outcomes of different perceptions can help identify in which aspects of feedback effectiveness generative AI needs improvement. This, in turn, can support the development of generative AI in a way that it can more effectively contribute to students' learning outcomes.

The findings of students evaluating human feedback more positively than feedback from AI might be explained by the actual quality of the feedback, type of task, or students' prior attitudes toward the feedback provider. Although prior research attempted to control for content of the feedback, it is still unclear whether students' perceptions of feedback relate to actual differences in quality of feedback between the sources. It remains an unanswered question whether students' perceptions are influenced by whom they believed the feedback came from. Gaining a deeper understanding is important, since it might indicate that generative AI is capable of providing feedback similar in quality to human feedback, but still be perceived as less effective once the source is identified as AI. One possible explanation for this might be trust in the source, as trust is recognized as an essential factor in accepting and adopting AI technology in various domains, including healthcare, autonomous transportation, data security and entertainment (Afroogh et al., 2024). Additionally, Bach et al. (2022) reported that people generally tend to have more trust in human agents than in AI systems. Considering this, it is important to explore how trust in AI systems might influence students' perceptions of feedback in educational contexts.

1.4. Trust and credibility

Trust reflects a willingness to accept and rely on a party, especially under conditions of uncertainty or vulnerability (Mayer et al., 1995). Bach et al. (2022) highlight that there is more than one way to define trust and that it should not be treated as a single, broad construct. Instead, researchers should focus on the most relevant components of trust within a specific context. In line with this, Shang et al. (2024) propose that trust in humans and AI systems is best understood through a two-dimensional model. This model includes affective trust (based on perceptions of benevolence and amiability) and cognitive trust (based on reliability, competence, understandability, and integrity). This approach allows for a more precise comparison of trust across these agencies.

Interestingly, Troy et al. (2024) found that trust is an important factor in the perception of feedback. Price et al. (2010) argue that trust in the feedback provider is necessary for students to be willing to use the feedback for improvement. Distrust in the feedback provider can reduce the effectiveness of feedback. Closely related to trust is the concept of credibility, which refers to both the perceived expertise and trustworthiness of the source (Liew & Tan, 2021). Schartel (2012) concludes that the credibility of the provider of feedback has an influence on the perception of feedback. Extending this, Van de Ridder et al. (2015) found that receiving feedback from a credible source may enhance feedback's acceptance, even when students find it less satisfactory. Taken together, these findings highlight the importance of including trust in the feedback provider as a potential factor explaining the differences of students' feedback perception between human and AI generated feedback.

Research has been done on credibility and trust in AI feedback sources. However, they provided different findings. Nazaretsky et al. (2024) tested for bias and found human feedback providers to be perceived as more credible than the AI feedback provider. In contrast, Abendschein et al. (2024) reported no significant differences in perceived credibility or trustworthiness between human and AI feedback providers. One reason for these conflicting findings may lie in the way how credibility and trust were measured. For example, Nazaretsky et al. (2024) assessed credibility only using three separate single-item constructs: reliability, ethics, and trust. In other words, their measurement of credibility was based on individual words representing different aspects rather than using a comprehensive scale consisting of multiple items. Similarly, Abendschein et al. (2024) did not account for the multiple dimensions of trust. These mixed findings may be the consequence of an oversimplified conception of trust, as it remains unclear whether different dimensions of trust varied within participants across the feedback sources. Additionally, the context in which the feedback was evaluated might also explain these conflicting results. In Abendschein et al.'s (2024) study, students evaluated the credibility of AI and human sources in a controlled setting without receiving feedback on their own work. The lack of personal relevance could have limited the ecological validity of the findings, as in real-world educational contexts, emotional and motivational factors often have an impact on how feedback is perceived. In contrast, this study considers the multi-dimensional model of trust, allowing for a more detailed understanding of which aspects of trust in a human versus generative AI might be most influential in shaping students' perceptions feedback effectiveness. Furthermore, this study aims for a more ecologically valid context, where participants receive feedback on their own reflective writing.

1.5. Influences on Trust

Trust in AI systems is influenced by multiple factors, including individual characteristics and specific system features. Individual factors such as education level, familiarity, and prior experience with AI play a significant role in influencing trust (Elkins & Derrick, 2013; Li et al., 2024). As AI tools become more integrated into educational contexts, addressing the conditions in which users trust these technologies is important for optimizing their potential (Tahiru, 2021).

Trust theories highlight the role of familiarity in building trust. According to Luhmann (1979, as cited in Gefen et al. 2003), familiarity helps individuals understand how someone or something behaves, which in turn facilitates the development of trust. Supporting this, Gefen (2000) and Zhu et al. (2023) found that greater familiarity and more frequent use of a system was associated with increased trust in that system.

In the specific case of AI chatbots, user trust may be influenced by factors such as reliability, security, and the transparency of the system's operations (Shahzad et al., 2024). Stöhr et al. (2024) found a positive correlation between familiarity with and usage of AI chatbots and users' positive attitudes toward these systems. This is particularly relevant for educational settings, where such positive effects might influence how students perceive feedback. However, the specific impact of familiarity on trust in AI chatbots remains underexplored. Given the increasing integration of AI chat-based systems in education, it is important to further explore how familiarity influences users' trust. Especially as this trust may play an important role in users' acceptance and perceived effectiveness of AI-generated feedback.

1.6. Mindset

Besides trust in the assessor, the mindset (fixed or growth) of a person receiving feedback shapes their acceptance and response to feedback (Forsythe & Johnson, 2016). Individuals with a fixed mindset tend to believe that abilities are stable traits that cannot be improved. People with a growth mindset, on the other hand, view abilities as changeable and capable of development through effort (Dweck, 2006).

This distinction is important in educational settings, since it affects how students perceive, interpret, and act on feedback. Forsythe & Johnson (2016) found that students with a growth mindset are more open to critical or challenging feedback and are more likely to use it to improve their performance. They tend to be more motivated to act on feedback and see it as a valuable part of the learning process. In contrast, students with a fixed mindset may feel threatened by feedback, distort it, or avoid it altogether.

Jeffs et al. (2021) found that mindsets can change over time with new experiences and targeted interventions. For example, they showed that feedback literacy training helps students shift to a growth mindset, which may improve their ability to engage with feedback effectively. This flexibility is particularly interesting because mindsets are domain specific, meaning that a person can have a fixed mixed mindset in one area (e.g., musicality) and a growth mindset in another (e.g., academic success; Alvarado et al., 2024).

However, it is still unclear how mindset interacts with perceptions of feedback when it is generated by AI. In general, AI-generated feedback tends to be evaluated more negatively than human feedback (Şık & Pektaş, 2024; Steiss et al., 2024). Since fixed mindset individuals are already more likely to avoid or distort feedback (Forsythe & Johnson, 2016), they may perceive AI-generated feedback as even less effective. However, this remains an assumption and hasn't been investigated yet. This highlights the importance of exploring how personal characteristics, such as mindset, interact with perceptions of feedback effectiveness generated by AI.

1.7. Research aims

Given the importance of receiving effective feedback on written reflections to support students' learning development and growth, the growing role of AI in education, and trust in the feedback provider, this study aims to investigate how the feedback provider (human vs. AI) influences students' perceptions of feedback effectiveness, with a key focus on the role of trust in the feedback provider. This results in the following research question:

> *How does the perceived feedback provider (human vs. AI) influence students' trust and perception of feedback effectiveness?*

In addition, the study will explore whether students' familiarity with AI chatbots and their frequency of use are associated with their trust in the AI feedback provider. Furthermore, the study includes exploratory analyses of the potential effect of correctly or incorrectly identifying the feedback provider, as well as the role of students' mindset.

Understanding these aspects can help identify which factors need improvement for implementing AI-generated feedback effectively in reflective writing.

2 Method

2.1. Experimental Design and Participants

The study used a within-subject experimental design with one manipulated factor: the perceived feedback (human vs. AI). This design isolates the effect of perceived feedback provider by keeping the content of the feedback constant across conditions. The experiment took place in the context of a reflective writing task, where feedback tends to be more subjective and personally relevant, with participants evaluating its effectiveness and their trust in the feedback provider. Beyond the main experimental manipulation, additional variables are participants' familiarity with the feedback provider, their guess of the source's identity, and mindset (fixed vs. growth).

Participants were recruited through a participant database and from the experimenter's network. An *a priori* power analysis was conducted to determine the required sample. Earlier work by Nazaretsky et al. (2024) examined the effect of feedback source (human vs. AI) on source credibility, reporting a large effect size (d = 0.88). To detect an effect size of Cohen's d = 0.88, a two-tailed Wilcoxon signed-rank test (matched pairs) was used, with α error = 0.05 and power = 0.9. This resulted in a required sample size of N = 17. However, to allow for additional analyses involving multiple within-subject measurements, a repeated-measures ANOVA (within factors) was also considered (f = 0.40, α error = 0.05, power = 0.9, two groups, five measurements). This resulted in a required sample size of N = 28.

Ultimately, a total of *N* = 39 participants took part in the study. Participants who failed an attention check were removed due concerns regarding the validity of their responses, which could compromise data reliability. Additionally, participants from the pilot study were excluded. This resulted in a final sample of *N* = 28 participants included in all analyses. All participants were studying at Eindhoven University of Technology, including nine bachelor students and 19 master students. Participants ranged in age from 19 to 26 years (*M* = 22.96, *SD* = 1.88), with an equal number of male and female participants.

2.2. Materials and Experimental Setup

The study took place in the Psychology Lab (PsyLab) of Eindhoven University of Technology. To enhance a sense of realism, a human assessor was physically present in the room and acted as the feedback provider. Participants individually completed a reflective writing task on a computer in one of the lab's cubicles. All communication between participants and the experimenter occurred through Discord.

The reflective writing task was adapted from a university-level course assignment. Based on a pre-test with six participants, a time limit of nine minutes was determined to be sufficient for completing the reflection task. To increase the credibility of the feedback process, participants were required to wait at least six minutes after submission to receive the feedback. This is based on the usual time the human assessor needs to provide thoughtful feedback.

The feedback participants received was generated through the use of an assessment rubric (see Appendix A), which was based on Gibbs' reflective cycle. To ensure that the feedback aligned with principles of formative and effective feedback, the prompt (shown in Figure 1) was designed according to the framework proposed by Hattie and Timperley (2007). The final prompt used in the AI chatbot was developed using the step-by-step approach outlined on the TU/e Prompt Engineering Guide website (TU Eindhoven AI Education, 2024). The prompt was tested and refined multiple times until it consistently produced relevant feedback responses that met the study's requirements. To ensure participants privacy and prevent any use of participants' reflections for AI training purposes, the prompt was entered into a secure Microsoft Copilot environment. Table 1 illustrates an example of the feedback generated by AI based on the reflection of a participant.

Figure 1

Prompt Template For AI Feedback Generation

Act as a Dutch University teacher. I would like you to give feedback on the reflective assignment of a student, directed to them. The feedback should focus only on the quality of their writing, including clarity, coherence, and mostly their personal depth of reflection, and adherence to the rubric criteria. Do not comment on the specific learning objective or the context of what they have learned. But mention personal things about their learning. It should be written as a coherent piece that's easy to read from start to finish. Don't use dash symbols in your generated text. The feedback you will give should be between 80 and 120 words. This was the reflective writing task: {reflective writing task} Use the following rubric: {assessment rubric} Based on this rubric, give feedback which is: 1. Is detailed, clear and specific and relates to specific areas of their work. 2. Tells students exactly how to improve. Some students also find it valuable if we tell them the reasons why the improvement is necessary. 3. Is honest and constructive. 4. Includes positive reinforcement. Praise for aspects of the work done well makes it easier for them to receive criticism. It also reinforces positive aspects of their work, explaining what it is they've done well and should keep doing. This was the answer of the student: {student answer}

Table 1

Sample AI Generated Feedback Based on a Participant's Reflective Writing

Assignment	Student response	Feedback from AI
Find one learning objective of a course you followed in the previous quartile, or you are following right now. Please write a reflective piece on your learning process and progress in relation to one of the course's learning objectives. Focus on a skill or area you engaged with and describe your experience in working with it. Try to include different aspects of your learning experience that you found relevant or meaningful.	Learning objective: Analyze and design computationally efficient algorithms for geovisualization processes. For a course I had last quartile I was supposed to learn to analyze and design efficient geovisualization programs, however, I felt that I did not make enough progress in this learning objective. I attended lectures, but as my background was not in computer science and algorithms, I struggled to follow them. As no reading materials were provided, I was often left trying to understand the material from the slides, which left gaps in my understanding of the material. The course had a project, where I worked with other students. This did help me a lot, as during the collaboration I could bounce ideas off the other students, deepening my understanding of the materials. Next to a group project, individual assignments were given, however, as no practice questions were provided, I did not feel confident in my initial attempts at the assignments. This feeling was correct, as I failed them. Altogether, this was a rather negative learning experience, as it felt like I had to already understand all course material to pass the course, without getting many opportunities to learn, get feedback or ask questions. In the end, I did not feel like I achieved the learning objective.	Your reflective assignment demonstrates a clear and honest account of your learning experience. You have effectively described the challenges you faced, particularly with your background in computer science and algorithms. Your reflection on the lack of reading materials and practice questions highlights a critical aspect of your learning environment. The collaboration with other students is a positive point, showing your ability to seek support and deepen your understanding through teamwork. To improve, consider providing more specific examples of how these challenges impacted your learning process. Additionally, offering a more detailed evaluation of your feelings and thoughts could enhance the depth of your reflection. Let me know once you're finished with reading the feedback.

2.3. Measures

Several measures were used to assess participants' perceptions and individual characteristics relevant to the study. These included perceived feedback effectiveness, trust in the feedback provider, familiarity with the feedback provider, and mindset (see Appendix B). An additional source-identification question (coded as: Turing Test; 0 = incorrect guess, 1 = correct guess) was included to indicate whether participants could correctly identify the origin of the feedback (human vs. AI), serving as a perception check for the manipulation. All participants received the measures in the same fixed order. However, within the AI condition questionnaire, which mirrored the human condition questionnaire, the order of the items was randomized to reduce potential order effects.

2.3.1 Perceived feedback effectiveness

Students' perceived effectiveness of feedback was measured by using the Feedback Perceptions Questionnaire (FPQ) developed by Strijbos et al. (2010) and the Feedback Effectiveness Scale (FES) developed by Lizzio and Wilson (2008).

In this study, the FPQ consisted of 18 items rated on a 7-point Likert scale ranging from fully disagree (1) to fully agree (7), to ensure consistency with the FES questionnaire and minimize potential confusion for participants. Answers to the 18 questions were divided into three scales. To measure students' motivation to take action and improve their performance, the mean scores of three items were calculated and used as the Willingness to Improve (W) score (α = .88). To measure whether the feedback meets students' learning needs, the mean scores of nine items were calculated to create the Perceived Adequacy of the Feedback (PAF) score (α = .89). To measure whether the feedback elicited positive emotions, the mean scores of six items were calculated to create an Affect (A) score (α = .86). Negatively phrased items were recoded so that the scale measured positive affect.

The FES was used to measure the contribution to students' development as learners. Although Lizzio and Wilson (2008) used this scale to measure overall feedback effectiveness, this study takes a different view. I argue that the FES only captures one part of feedback effectiveness, namely how much the feedback helped students grow as learners. Therefore, in this study, the FES is seen as measuring just that specific aspect, not overall effectiveness. The scale consisted of three items that were rated on a 7-point Likert scale to measure the contribution to students' development as learners (1 = not at all, 7 = very). The mean scores of the three items were calculated to create a Feelings of Empowerment (FE) score (α = .63).

2.3.2 Trust

To measure trust in AI, the semantic differential scale developed by Shang et al. (2024) was used in the study. This scale captured both affective and cognitive dimensions of trust through pairs of opposing adjectives (e.g., unreliable–reliable, apathetic–empathetic). The scale was adapted to reduce the length of the questionnaire due to time constraints. The selection of items for removal is based primarily on their factor loadings. While all items exceed the commonly accepted threshold of .60, preference was given to loadings above .70 to ensure the scale would still maintain good quality even if fewer items were retained. The final questionnaire consisted of 20 items that were rated on a 5-point response format. Responses to the 20 questions were divided into a Cognitive Trust (CT) score, which consisted of 14 items (α = .89), and an Affective Trust (AT) score, which consisted of six items (α = .79). In addition, one single-item questionnaire adapted from Yin et al. (2019) was included to assess general trust in the assessor (1 = no trust at all, 5 = full trust).

2.3.3 Familiarity with the feedback provider

To assess participants' familiarity with and frequency of use of chat-based AI, this study adapted single-item questions based on the questionnaire used by Elhassan et al. (2025). Participants were asked how familiar they were with chat-based AI software (AI Familiarity), with response options ranging from "Not familiar at all" to "Extremely familiar" on a scale from 0 to 4. Frequency of AI chatbot use was measured by two questions: how often participants used chat-based AI for educational purposes (AI Frequency) and how often they asked chat-based AI to provide feedback on their written work (AI Feedback Use). Both questions used the response options: “Never,” “Rarely,” “Monthly,” “Weekly,” and “Daily,” also on a scale from 0 to 4.

Familiarity with the assessor was measured using two items. The first item asked participants to indicate the extent to which they knew the assessor (Human Familiarity), with response options ranging from “I don’t know him/her” to “Extremely well” on a scale from 0 to 4. The second item assessed prior contact with the assessor by asking whether the participant had the assessor as a teacher in the previous quartile (Q3), the current quartile (Q4), both, or neither. These responses were later recoded into a binary variable indicating whether the participant had the assessor as a teacher during either period (coded as: Teacher; 0 = did not have the assessor as a teacher in Q3 or Q4, 1 = had the assessor as a teacher in Q3 or Q4).

2.3.4 Mindset

To measure individuals’ mindset, the Multidimensional Mindset Scale (MUMIS) developed by Alvarado et al. (2024) was used in this study. This was chosen because it provided a more comprehensive and refined measurement than Dweck’s original scale. Unlike the unidimensional model, this scale treated growth and fixed mindsets as independent constructs, allowing individuals to hold both types of beliefs across different dimensions (i.e., intelligence beliefs, practice and effort, challenges, and multiple intelligences). The questionnaire included 25 items that were rated on a 6-point Likert scale ranging from strongly disagree (1) to strongly agree (6). Answers to the 25 questions were divided into two scales. The Fixed Mindset score, which consisted of 14 items (α = .70), and the Growth Mindset score, which consisted of 11 items (α = .77).

2.4. Procedure

Prior to data collection, the study was approved by the Ethical Review Board at Eindhoven University of Technology.

Upon arrival in the lab, participants read and signed the informed consent form (Appendix C). They received the following instructions:

> “*Today, you'll write a reflection and receive feedback on this written reflection from an instructor who is an expert in assessing student self-reflections and has extensive experience in this area. We’re interested in understanding how you experience this feedback and what you think about it.*
>
> *Different instructors are involved, and the instructor present today is [name of the assessor]. To ensure your privacy and anonymity, I will be the only point of contact - both for you and for [name of the assessor]. I have a separate channel/chat in Discord with you than with [name of the assessor]. [Name of the assessor] will never know who wrote what, and he/she won’t have access to any of your questionnaire responses before they are anonymized. Please try to be as honest as possible, as your genuine thoughts will help us understand your experience better.*
>
> *I will now ask you to get to your seat and follow the instructions in Discord. If you have any questions during the study, please stay in your designated seat and send your questions to me via Discord. Do you have any questions right now?*
>
> *In the cubicle you will find the informed consent form, please read it first and sign. Then you can start the experiment. There is also a small piece of paper that provides your participant*

number.

Please leave your bags and phone outside of the cubicle. "

The study consisted of five steps (shown in Figure 2). The first step consisted of participants reading the message that was send in the Discord, which was the following:

"*Hello, and thank you for participating in this study! Your participant number is: [insert number]. Remember this study is completely voluntary, and you're free to stop at any time - just let me know via the Discord chat. If you have any questions during the process, feel free to send them here as well.*
Also, if it seems like I'm taking a bit too long to respond, feel free to tag me with @researcher. This helps me get a clearer notification, so I can respond to you more quickly!

You will start by writing a short reflection task, for which you will have a time limit of 9 minutes. You can write your reflection in Word so you can edit it freely. Once you're finished, copy and paste your final text into this Discord chat. I'll send you a 2-minute warning before the time is up, so you have time to wrap things up. Please send a message in Discord when you're ready to start."

Once the participants responded, the following message was sent:

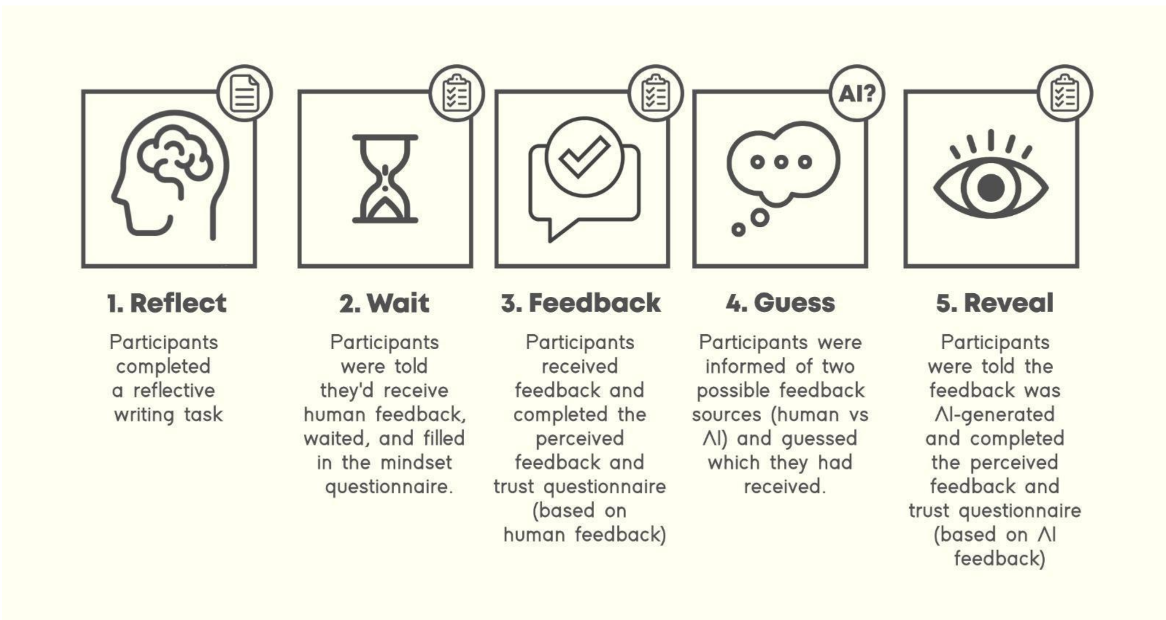
"Think of one learning objective (usually phrased as: "after the course, students are able to …") of a course you followed in the previous quartile, or you are following right now.

Please write a reflective piece on your learning process and progress in relation to this learning objective. Try to include different aspects of your learning experience that you found relevant or meaningful.

Good luck!"

Figure 2

Overview of The Five-step Experimental Procedure



Then the participants performed the reflective writing task based on a course they were currently taking or had recently completed and were instructed to complete the task within 9 minutes. Reminder messages were sent after 7 and 9 minutes, respectively, if the participant had not yet submitted their reflection.

“Just a heads-up! You have 2 minutes left to finish your reflection. Feel free to start wrapping things up. If you’re already done, you’re welcome to send it in now!”

“It’s absolutely fine if your reflection isn’t completely finished. However, the time is up, so please send me what you’ve written so far.”

During the second step, participants were informed that they would need to wait briefly as the assessor would require time to evaluate their written reflection. In reality, the feedback provided was always generated by an AI system. In the meantime, they completed the first questionnaire (mindset). The exact message in Discord was the following:

“Thank you for sending your reflection. You’ve been selected to receive feedback first. Please wait while the assessor reviews your work and prepares feedback - this will take approximately 4 to 6 minutes.

In the meantime, please complete the following questionnaire: [Insert link to mindset questionnaire]
Once you’re finished, please return to this Discord chat and send me a message to let me know. Thanks!”

Once the participants responded before 6 minutes, the following message was sent:

“The assessor is still reviewing your reflection, so it will take a little bit more time. Please wait a little longer to receive your feedback.“

The third step included receiving the feedback and filling in the second questionnaire aimed at evaluating the effectiveness of the feedback, their trust in the assessor, and their perceived familiarity with the assessor. The participants received this message in Discord:

“The assessor finished reviewing your work!
Please take a moment to read the feedback you received on your reflection task:
[Insert feedback]
Let me know once you’re finished with reading the feedback.”

Once the participants responded, the following message was sent:

“Please complete the following questionnaire: [insert link of trust and perception of feedback questionnaire of the assessor]
Once you’re finished, please return to this Discord chat and send me a message to let me know. Thanks!”

While participants completed the questionnaire, the feedback was deleted from the Discord chat, preventing them from reviewing it again.

In the fourth step the participants were informed that there were two conditions: one in which the feedback was written by a human assessor and another in which it was generated by AI. They were then asked to indicate which condition they believed they were in. They were subsequently informed that the feedback had been generated by AI, as it was in all cases. The following message was send in the chat:

"Thank you for filling in the questionnaire!

I have to tell you something. This study actually includes two different conditions: one in which the feedback is written by the human assessor and another in which it is generated by an AI tool. Half of the participants received feedback from a human, and the other half received AI-generated feedback.

In which condition do you think you were placed? Please let me know in the chat what you think."

In the fifth and last step, the participants were asked to review the feedback once more. They received the following message in Discord:

"In your case, the feedback was actually/indeed generated by an AI tool.
Please take a moment to read the feedback again:

[Insert feedback]

Let me know once you're finished with reading the feedback."

Once the participant responded they were asked to complete the third questionnaire in order to determine whether their ratings would change now that they knew the feedback was AI-generated. The question regarding their familiarity with the assessor was replaced with questions about their familiarity and frequency use of AI. They received the following message:

"Please complete the following questionnaire: [insert link of trust and perception of feedback questionnaire of the AI assessor]
Once you're finished, please return to this Discord chat and send me a message to let me know. Thanks!"

Once the final questionnaire was completed, participants were debriefed and informed of the true purpose of the study. The last message they received on Discord was:

"Thank you for your answers and for participating in this experiment! You are now finished. I will come to get you shortly."

At the end of the session, participants were thanked for their participation and received compensation.

2.5. Statistical Analysis

All statistical analyses were conducted using Stata/BE 18.0. Prior to running any analysis, the data was cleaned and examined for outliers. Following that, relevant assumptions for the analyses, including normality, homoscedasticity, linearity, and multicollinearity, were checked. Most assumption checks showed no significant violations and are therefore not discussed further. Assumptions that were not fully met, however, are addressed where relevant.

To investigate whether the perceived feedback provider (human vs. AI) influenced participants' perceptions of feedback effectiveness and trust in the source, a one-way repeated-measures multivariate analysis of variance (MANOVA) was conducted. To further explore which specific dependent variables contributed to this overall effect, follow-up repeated-measures ANOVAs were conducted separately for each construct. The main predictor was Condition (0 = human, 1 = AI), and independent outcome variables were General Trust (GT), Affective Trust (AT), Cognitive Trust (CT), Willingness to Improve (W), Perceived Adequacy (PA), Feedback Affect (FA), and Feelings of Empowerment (FE).

Furthermore, to explore whether trust explains the relationship between the perceived feedback provider and the perceived feedback effectiveness scores, mediation models with repeated measures were conducted. Each model tested whether one of the three types of trust acted as a mediator between the conditions and one of the feedback effectiveness outcome variables. This resulted in a total of twelve separate mediation models, allowing for a clear examination of how each type of trust uniquely mediates the effect of the two conditions on the different types of feedback effectiveness perceptions. Indirect effects were tested using nonparametric bootstrapping with 1000 resamples and 95% confidence intervals. All mediation analyses included Condition as the independent variable and controlled for age, gender (0 = female, 1 = male), and education level (0 = bachelor student, 1 = master student). A random intercept for participants was included to account for the within-subjects design (56 observations across 28 participants).

Finally, exploratory analyses were conducted. To examine whether trust in the feedback provider was influenced by prior experience with either AI or the human assessor, multiple linear regression analyses were conducted separately within each condition. All continuous predictors (AI Familiarity, AI Frequency, AI Feedback Use, Human Familiarity) were mean-centered, and covariates were included in the models.

To assess whether individual differences in AI experience predicted participants' ability to correctly identify the source of the feedback, a binary logistic regression was conducted including covariates and mean-centered continuous predictors. In addition, to test whether source identification moderated the effect of the two conditions on feedback effectiveness and trust scores, mixed-effects models were conducted. Once again, all continuous predictors were mean-centered, all models included covariates, and in addition a random intercept for participants was included.

Lastly, to explore the influence of mindset, participants were categorized into four mindset groups using median splits on the growth and fixed mindset scales.

As the covariates age, gender, and education level did not substantially alter the main effects and were not significant (all *ps* $> .150$), they are not discussed in detail in the Results section. Furthermore, all statistical tests used a significance threshold of $p < .05$.

3 Results

The following section presents the results of the analyses conducted to test the effects of perceived feedback source (human vs. AI) on participants' trust and perceptions of feedback effectiveness. First, the direct effects of the perceived feedback source on the outcome variables are examined. Afterwards, a series of mediation analyses explore whether trust mediates these effects. In addition, exploratory analyses investigate individual differences, including participants' familiarity with the feedback provider and their ability to correctly identify the provider of the feedback. Finally, the role of students' mindset was considered.

3.1. Effect of Perceived Agency on Feedback Perception and Trust

To examine whether the perceived source of feedback (human vs. AI) influenced participants' trust and perceptions of feedback effectiveness, a one-way repeated measures MANOVA was conducted. The model included seven dependent variables: Willingness to Improve (W), Perceived Adequacy of Feedback (PAF), Affect (A), Feelings of Empowerment (FE), General Trust (GT), Cognitive Trust (CT), and Affective Trust (AT). A significant multivariate effect of Condition was found, Wilks' Lambda = .336, $F(7, 21) = 5.93$, $p < .001$, indicating a difference in the level of participants' feedback effectiveness and trust scores between the two conditions. Follow-up univariate tests, specifically repeated-measures ANOVA, further demonstrated significant effects of Condition on all dependent variables (see Table 2). Minor violations of normality were observed for W, A, and CT (significant kurtosis and joint skewness-kurtosis tests; all *ps* < .05). Nonetheless, Wilcoxon signed-rank tests confirmed the robustness of the effects, with all effects significant (all *ps* < .05).

Participants reported significantly higher W scores after receiving feedback they believed came from a human compared to feedback labeled as AI-generated, $F(1, 27) = 34.35$, $p < .001$, $\eta^2 = .56$. Similarly, participants reported higher PAF scores in the human condition than in the AI condition, ($F(1, 27) = 16.39$, $p < .001$, $\eta^2 = .38$. Participants also reported higher A scores in the human condition than in the AI condition, $F(1, 27) = 6.75$, $p = .015$, $\eta^2 = .20$. Additionally, participants reported higher FE scores as well in the human condition compared to the AI condition, $F(1, 27) = 7.68$, $p = .010$, $\eta^2 = .22$. These differences are illustrated in Figure 3, which displays the mean ratings across the feedback effectiveness dimensions for both feedback providers.

Regarding trust, participants also reported higher GT scores for the human feedback provider than the AI feedback provider, $F(1, 27) = 21.77$, $p < .001$, $\eta^2 = .45$. Similarly, participants rated CT scores higher in the human condition than in the AI condition, $F(1, 27) = 9.79$, $p = .004$, $\eta^2 = .27$. Finally, participants also rated AT scores higher in the human condition than in the AI condition, $F(1, 27) = 14.29$, $p = .001$, $\eta^2 = .35$. Figure 4 shows these trust differences between the two feedback sources.

3.2. Mediation Analysis: Trust as a Mediator

To examine whether trust mediated the relationship between the perceived feedback source and perceived feedback effectiveness, a series of mixed-effects mediation models were estimated using bootstrapping with 1000 resamples (see Methods: Statistical Analysis). Feedback effectiveness was measured using four outcome variables: W, PAF, A, and FE.

Table 3 shows the correlations among variables for the human condition, while Table 4 shows the correlations for the AI condition.

The interaction effect of Condition on the relationship between any trust measure and any feedback effectiveness measure was not significant in any case (all *ps* > .478), indicating that the effect of trust on feedback effectiveness did not differ depending on whether the feedback provider is perceived as human or AI.

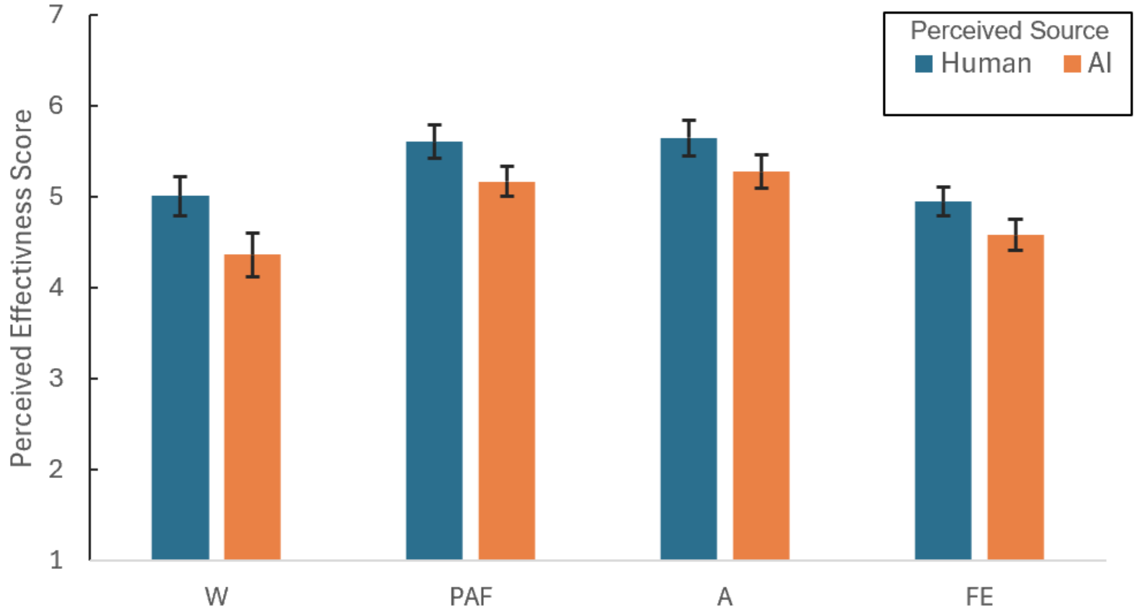
Table 2

Feedback Effectiveness and Trust: Human vs. AI Conditions

Measure	Scale	Human		AI		$F(1, 27)$	p	η^2
		M	SD	M	SD			
Willingness to Improve (W)	(1-7)	5.01	1.16	4.37	1.26	34.35	< .001	.56
Perceived Adequacy of Feedback (PAF)	(1-7)	5.61	0.95	5.17	0.85	16.39	< .001	.38
Affect (A)	(1-7)	5.65	1.04	5.28	0.98	6.75	.015	.20
Feelings of Empowerment (FE)	(1-7)	4.95	0.84	4.58	0.91	7.68	.010	.22
General Trust (GT)	(1-5)	4.04	0.79	3.32	0.67	21.77	< .001	.45
Cognitive Trust (CT)	(1-5)	4.08	0.58	3.84	0.53	9.79	.004	.27
Affective Trust (AT)	(1-5)	3.78	0.55	3.51	0.65	14.29	.001	.35

Figure 3

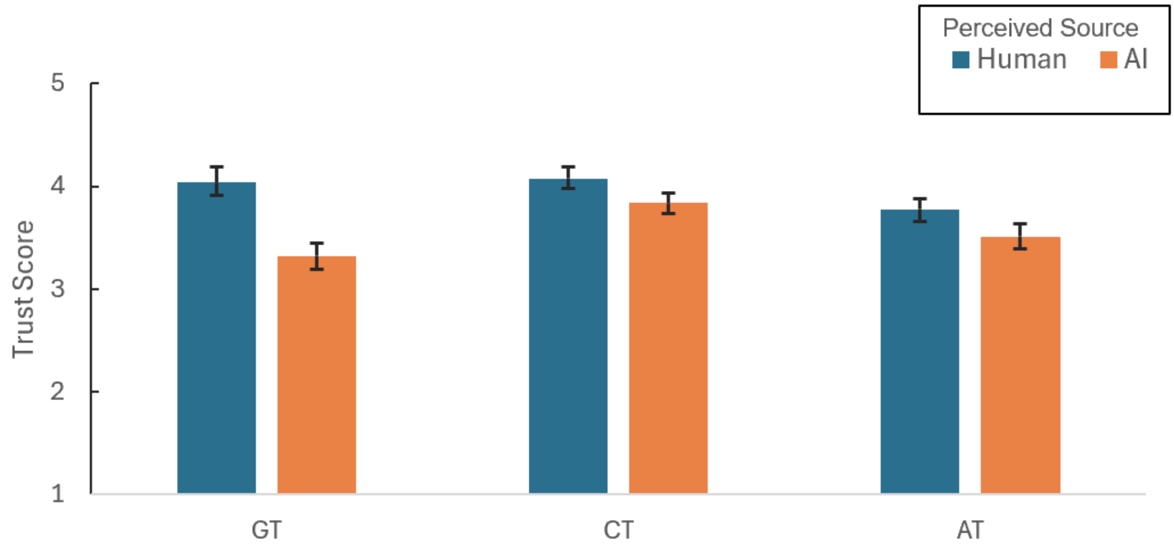
Human Feedback is Perceived as More Effective Than AI-generated Feedback



Note. Visualization of average values on perceived feedback effectiveness scores across four dimensions: Willingness to Improve (W), Perceived Adequacy of Feedback (PAF), Affect (A), and Feelings of Empowerment (FE), separately for the human and AI feedback conditions. Error bars indicate standard errors.

Figure 4

Human Feedback Provider is Trusted More Than AI Feedback Provider



Note. Visualization of average values on trust scores across three dimensions: General Trust (GT), Cognitive Trust (CT), and Affective Trust (AT), separately for the human and AI feedback conditions. Error bars indicate standard errors.

Table 3

Pairwise Correlations Between the Variables in the Human Condition

Variables	1	2	3	4	5	6	7
1. GT	-						
2. CT	.72***	-					
3. AT	.53**	.75***	-				
4. W	.25	.47*	.47*	-			
5. PAF	.65***	.74***	.54**	.35	-		
6. A	.41*	.66***	.53**	.35	.67***	-	
7. FE	.37	.61***	.58**	.27	.67***	.61***	-

Note. N = 28. GT = General Trust; CT = Cognitive Trust; AT = Affective Trust; W = Willingness to Improve; PAF = Perceived Adequacy of Feedback; A = Affect; FE = Feelings of Empowerment. * $p < .05$. ** $p < .01$. *** $p < .001$.

Table 4

Pairwise Correlations Between the Variables in the AI Condition

Variables	1	2	3	4	5	6	7
1. GT	-						
2. CT	.60***	-					
3. AT	.49**	.79***	-				
4. W	.28	.36	.50**	-			
5. PAF	.66***	.79***	.74***	.52**	-		
6. A	.31	.49**	.55**	.55**	.71***	-	
7. FE	.27	.56**	.37	.37	.72***	.58**	-

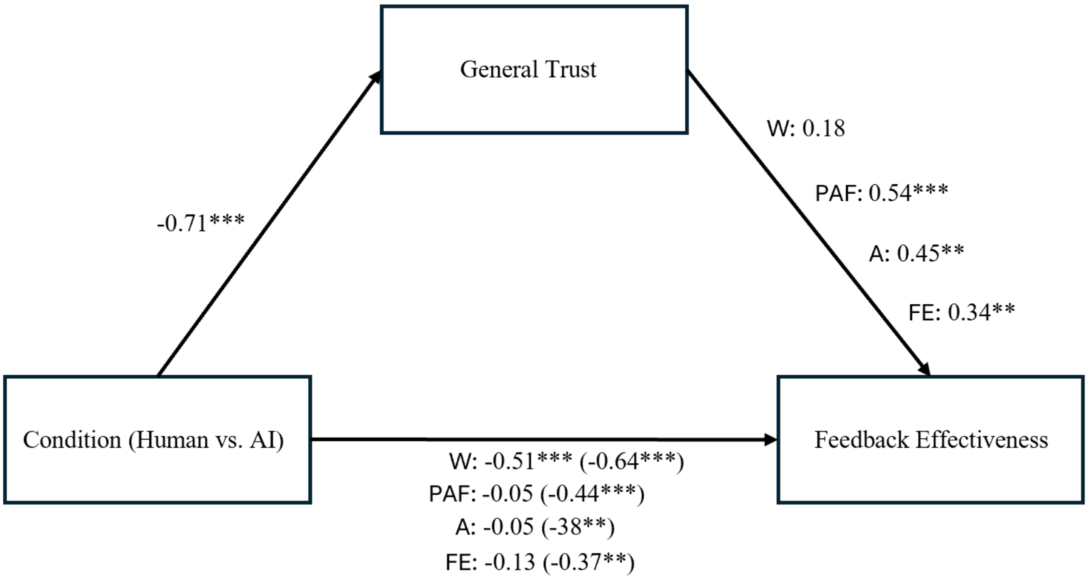
Note. N = 28. GT = General Trust; CT = Cognitive Trust; AT = Affective Trust; W = Willingness to Improve; PAF = Perceived Adequacy of Feedback; A = Affect; FE = Feelings of Empowerment. * $p < .05$. ** $p < .01$. *** $p < .001$.

3.2.1 Mediation via General Trust

To examine whether GT mediated the relationship between Condition and feedback effectiveness, mediation analyses were conducted for two outcome variables: PAF and A (see Figure 5). Results showed a significant total effect of Condition on GT (b = -0.71, p < .001), with lower general trust in the AI feedback provider compared to the human feedback provider. GT was positively associated with PAF, A and FE, meaning higher general trust leads to higher feedback effectiveness scores (in terms of perceived adequacy of the feedback, affective responses, and feelings of empowerment). Significant total effects of Condition were found for both outcomes, meaning that the source of feedback influenced the outcomes of feedback effectiveness. Importantly, GT mediated these relationships. Significant indirect effects via GT were found for PAF (b = -0.39, 95% CI [-0.56, -0.22], p < .001), A (b = -0.32, 95% CI [-.544, -.098], p = .005), and FE (b = -0.24, 95% CI [–0.41, –0.08], p = .004). The direct effects of Condition on these outcomes were no longer significant after accounting for GT, indicating full mediation. This suggests that the differences in PAF, A and FE scores between the human and AI conditions are explained by the level of general trust participants have in the feedback source. In contrast, the indirect effect via GT of Condition on W was not significant (b = -0.13, 95% CI [–0.27, 0.01], p = .061), and the direct effect remained significant (b = -0.51, 95% CI [–0.77, –0.25], p < .001). This suggests that GT did not mediate the effect of Condition on W. In other words, the different scores of willingness to improve across the conditions were not explained by general trust.

Figure 5

Mediation Effects of General Trust on the Relationship Between Condition and Feedback Outcomes



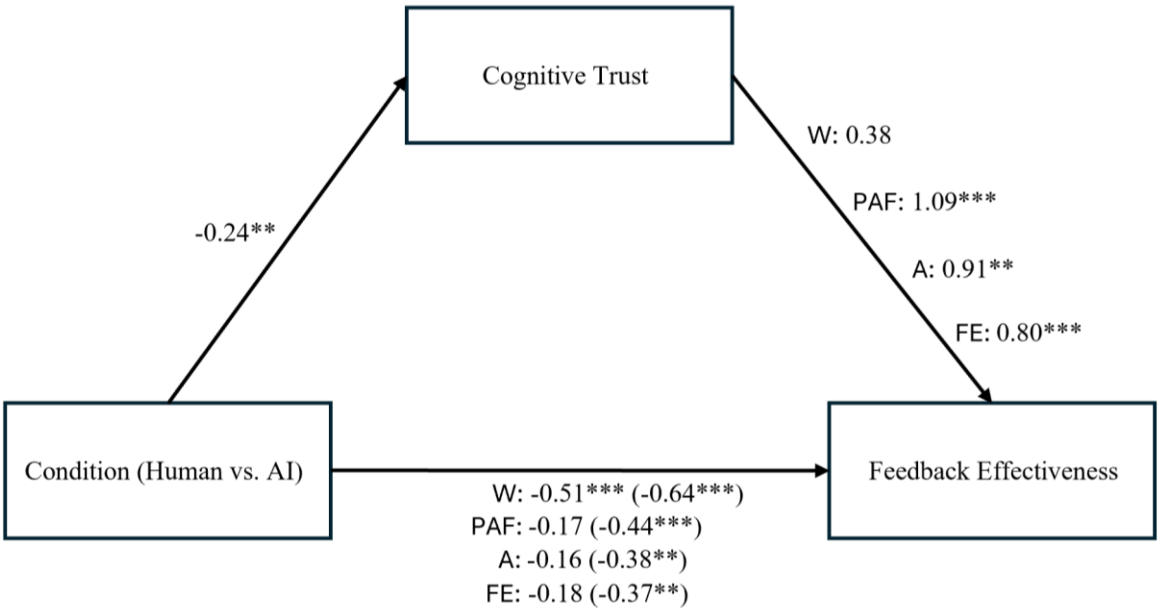
Note. Mediation analysis path diagram with direct effects, and with the total effect added in the parenthesis. All coefficients represent unstandardized regression coefficients. The negative coefficients indicate lower scores in the AI condition compared to the human condition. Feedback effectiveness was assessed using four outcomes: Willingness to Improve (W), Perceived Adequacy of Feedback (PAF), Affect (A), and Feelings of Empowerment (FE). Random intercepts for participants were included in all models to account for repeated measures. **p < .01. ***p < .001.

3.2.2 Mediation via Cognitive Trust

To examine whether CT mediated the relationship between Condition and each of the four outcome variables: W, PAF, A, and FE, mediation analyses were conducted (see Figure 6). Results showed a significant total effect of Condition on CT (*b* = -0.24, *p* = .001), with lower cognitive trust in the AI feedback provider compared to the human feedback provider. The effect of CT on feedback effectiveness was positive and significant for PAF, A, and FE, meaning higher cognitive trust leads to higher scores of feedback effectiveness (in terms of perceived adequacy of the feedback, affective responses and feelings of empowerment). Significant total effects of Condition were found for all four outcomes, indicating that the feedback source influenced feedback effectiveness measures. Importantly, CT mediated the relationships with PAF, A, and FE. Significant indirect effects via CT were found for PAF (*b* = -0.26, 95% CI [–0.42, –0.10], *p* = .001), A (*b* = -0.22, 95% CI [–0.43, –0.01], *p* = .038), and FE (*b* = -0.19, 95% CI [–0.35, –0.04], *p* = .012). After accounting for CT, the direct effects of Condition on these outcomes were no longer significant, indicating full mediation. This indicates that the differences in PAF, A, and FE scores between the two conditions are explained by the level of cognitive trust participants have in the feedback provider. In contrast, the indirect effect via CT of Condition on W was not significant (*b* = -0.09, 95% CI [–0.26, 0.07], *p* = .265), and the direct effect remained significant (*b* = -0.55, 95% CI [–0.82, –0.28], *p* < .001). This suggests that CT did not mediate the effect of Condition on W. In other words, the different scores of willingness to improve across the conditions were not explained by cognitive trust.

Figure 6

Mediation Effects of Cognitive Trust on the Relationship Between Condition and Feedback Outcomes



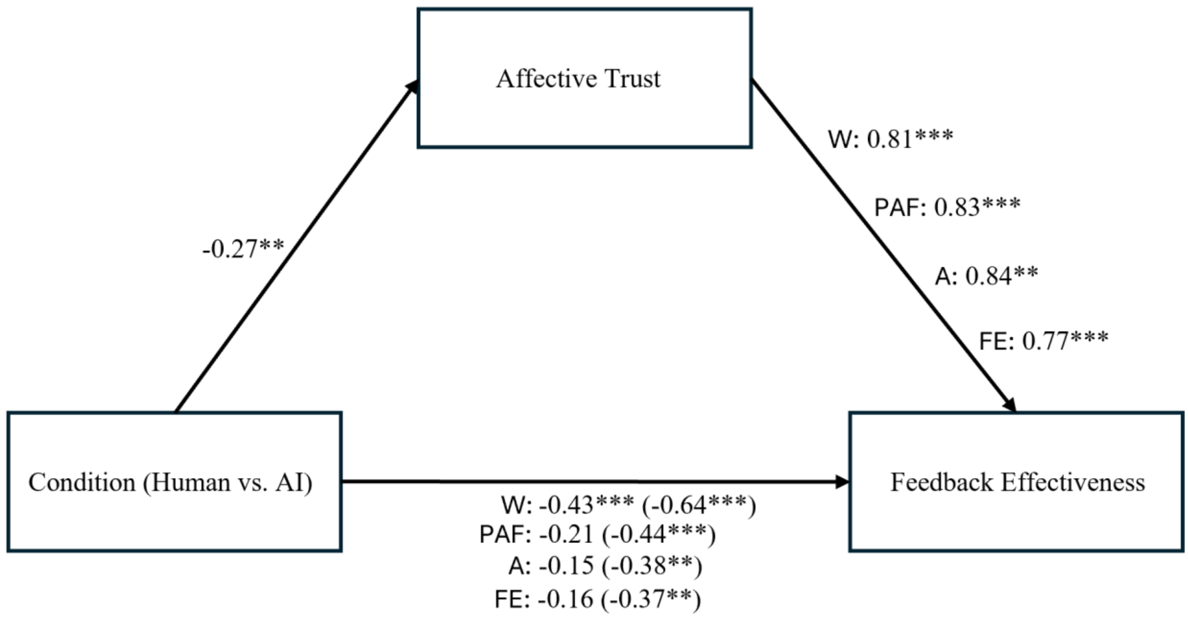
Note. Mediation analysis path diagram with direct effects, and with the total effect added in the parenthesis. All coefficients represent unstandardized regression coefficients. The negative coefficients indicate lower scores in the AI condition compared to the human condition. Feedback effectiveness was assessed using four outcomes: Willingness to Improve (W), Perceived Adequacy of Feedback (PAF), Affect (A), and Feelings of Empowerment (FE). Random intercepts for participants were included in all models to account for repeated measures. **p < .01. ***p < .001.

3.2.3 Mediation via Affective Trust

To examine whether AT mediated the relationship between Condition and the four outcome variables, mediation analyses were conducted (see Figure 7). Results showed a significant total effect of Condition on (*b* = -0.27, *p* < .001), with lower affective trust in the AI feedback provider compared to the human feedback provider. AT was positively associated with all four outcomes, indicating that higher affective trust leads to higher feedback effectiveness scores (in terms of willingness to improve, perceived adequacy of the feedback, affective responses, and feelings of empowerment). Significant total effects of Condition on all four outcomes were significant, indicating that the source of feedback influenced these measures of feedback effectiveness. Importantly, AT significantly mediated the relationship between Condition and all four outcomes. Significant indirect effects via AT were found for W (*b* = -0.22, 95% CI [–0.37, –0.07], *p* = .005), PAF (*b* = -0.22, 95% CI [–0.36, –0.08], *p* = .002), A (*b* = -0.23, 95% CI [–0.39, –0.06], *p* = .009), and FE (*b* = -0.21, 95% CI [–0.36, –0.05], *p* = .008). For PAF, A, and FE, the direct effects of Condition were no longer significant, indicating full mediation via AT. This suggests that differences in these feedback effectiveness scores between the human and AI conditions can be explained by differences in affective trust. However, for W, both the indirect effect (*b* = -0.22, 95% CI [–0.37, –0.07], *p* = .005) and the direct effect (*b* = -0.43, 95% CI [–0.69, –0.17], *p* = .001) remained significant, indicating partial mediation. In other words, affective trust partly explains the difference in willingness to improve across conditions.

Figure 7

Mediation Effects of Affective Trust on the Relationship Between Condition and Feedback Outcomes



Note. Mediation analysis path diagram with direct effects, and with the total effect added in the parenthesis. All coefficients represent unstandardized regression coefficients. The negative coefficients indicate lower scores in the AI condition compared to the human condition. Feedback effectiveness was assessed using four outcomes: Willingness to Improve (W), Perceived Adequacy of Feedback (PAF), Affect (A), and Feelings of Empowerment (FE). Random intercepts for participants were included in all models to account for repeated measures. **p < .01. ***p < .001.

3.3. Exploratory Analyses

3.3.1 Familiarity in The Feedback Provider

To explore whether students' familiarity and frequency of interaction with AI chatbots were related to their trust in the AI feedback provider, multiple linear regression analyses were conducted (Condition = 1; see Methods: Statistical Analysis). Additionally, to explore whether familiarity with the human assessor or having the assessor as a teacher influenced levels of trust, multiple linear regression analyses were conducted as well (Condition = 0).

Before conducting the regression analyses, descriptive statistics for the main predictors were computed. Participants reported relatively high scores on AI Familiarity (M = 3.04, SD = 0.64), as well as for AI Frequency (M = 3.46, SD = 0.69). For AI Feedback Use (M = 2.36, SD = 0.99) participants reported moderate scores. Furthermore, the descriptive statistics showed that on average, participants reported low familiarity with the assessor (M = 1.39, SD = 1.17) scores, and 21% of the participants indicated that they had the assessor as a teacher. Next, the regression analyses were performed.

For GT in the AI condition, the model was not statistically significant, $F(6, 21) = 0.49$, $p = .810$, $R^2_{adj} = -0.13$. None of the main predictors, nor the covariates, significantly predicted GT. Specifically, AI Familiarity (b = -0.44, p = .138), AI Frequency (b = 0.30, p = .266), and AI Feedback Use (b = 0.03, p = .875) showed non-significant relationships with GT.

For CT in the AI condition, the model was also not significant, $F(6, 21) = 0.89$, $p = .523$, $R^2_{adj} = -0.03$. None of the predictors significantly predicted CT. Specifically, AI Familiarity (b = -0.35, p = .119), AI Frequency (b = 0.18, p = .375), and AI Feedback Use (b = -0.12, p = .315) showed non-significant relationships with CT.

For AT in the AI condition, the model was once again not statistically significant, $F(6, 21) = 0.20$, $p = .973$, $R^2_{adj} = -0.22$. None of the predictors significantly predicted AT. Specifically, AI Familiarity (b = -0.22, p = .444), AI Frequency (b = 0.14, p = .591), and AI Feedback Use (b = -0.07, p = .673) showed non-significant relationships with AT.

Overall, these results indicate that participants' prior experience with AI, in terms of familiarity and frequency of use, did not significantly influence their reported level of trust in the AI feedback provider. Although the negative coefficient for AI Familiarity suggests that greater familiarity with AI may be associated with slightly lower trust in the AI feedback provider, this effect was not statistically significant and should therefore be interpreted with caution.

For GT in the human condition, the overall model was not significant, $F(5, 22) = 1.48$, $p = .236$, $R^2_{adj} = 0.08$. Human Familiarity had no significant effect on GT (b = -0.33, p = .055). In contrast, Teacher was significantly associated with higher GT (b = 1.09, p = .021), indicating that having the assessor as a teacher may positively impact students' general trust in the assessor.

For CT in the human condition, the model showed no significance, $F(5, 22) = 2.33$, $p = .077$, $R^2_{adj} = 0.20$. Human familiarity did not have a significant effect on CT (b = -0.18, p = .124). However, Teacher showed a positive significant effect on CT in the human condition (b = 0.86, p = .009), indicating that having the assessor as a teacher may positively impact students' cognitive trust in the assessor.

For AT in the human condition, the model was also not significant, $F(5, 22) = 1.02$, $p = .429$, $R^2_{adj} = 0.004$. Neither Human Familiarity (b = -0.04, p = .722) nor Teacher (b = 0.47, p = .155) showed a significant effect on AT in the human condition. This suggests that participants familiarity levels in the human assessor or having the assessor as a teacher did not affect students' affective trust in the assessor.

3.3.2 Turing Test

After participants evaluated the feedback in the human condition, they were informed that the feedback could have been provided by an AI. To explore whether participants were able to correctly identify the source of the feedback provider, a source-identification item was included. Of the 28 participants, 21 correctly identified the feedback as AI-generated, while 7 believed it was written by a human.

To examine whether individual differences in AI experience predicted accurate source identification, a logistic regression analysis was conducted (see Methods: Statistical Analysis) with AI Familiarity, AI Frequency, and AI Feedback Use as predictors. The overall model was not statistically significant, $\chi^2(6, N = 28) = 7.21$, $p = .30$, indicating that these variables did not reliably predict Turing Test. None of the individual predictors achieved statistical significance. Specifically, AI Familiarity was not a significant predictor for Turing Test ($OR = 2.365$, 95% CI [0.30, 23.2], $p = .379$), neither was AI Frequency ($OR = 2.15$, 95% CI [0.34, 13.5], $p = .412$), nor was AI Feedback a significant predictor for Turing Test ($OR = 0.41$, 95% CI [0.09, 1.99], $p = .270$). This suggests that individual differences in prior experience with AI chatbots did not significantly influenced participants' ability to correctly identify the source of the feedback.

To further explore the role of source identification accuracy, additional analyses were conducted to examine whether there was a difference in feedback effectiveness and trust outcomes between the conditions whether participants correctly or incorrectly guessed the feedback source. Separate mixed-effects regression models were estimated (see Methods: Statistical Analyss) for W, PAF, A, FE, GT, CT, and AT.

For none of the feedback effectiveness scores as well as the trust scores was there a significant interaction effect between Condition and Turing Test; W ($b = 0.41$, 95% CI [–0.05, 0.88], $p = .081$), PAF ($b = 0.31$, 95% CI [–0.16, 0.77], $p = .196$), A ($b = 0.42$, 95% CI [–0.20, 1.04], $p = .185$), FE ($b = –0.11$, 95% CI [–0.70, 0.48], $p = .712$), GT ($b = 0.19$, 95% CI [–0.49, 0.87], $p = .581$), CT ($b = 0.18$, 95% CI [–0.16, 0.52], $p = .295$), and AT ($b = –0.01$, 95% CI [–0.32, 0.31], $p = .961$). This indicates that Turing Test did not significantly moderate the effect of Condition on any of the feedback effectiveness scores nor the trust scores. In other words, participants' feedback effectiveness and trust scores between the two conditions did not differ depending on their accuracy in guessing the feedback source.

3.3.3 Role of Mindset

To examine whether students' mindsets influenced their perception of the feedback depending on the perceived feedback provider, participants were first categorized into four mindset groups. This categorization was based on a median split of both the fixed and growth mindset scales (see Methods: Statistical Analysis). However, the two medians are not necessarily independent, and individuals may simultaneously hold both types of beliefs to varying degrees. Specifically, individuals could score above the median on one variable while score below the median on the other, leading to variations in group sizes.

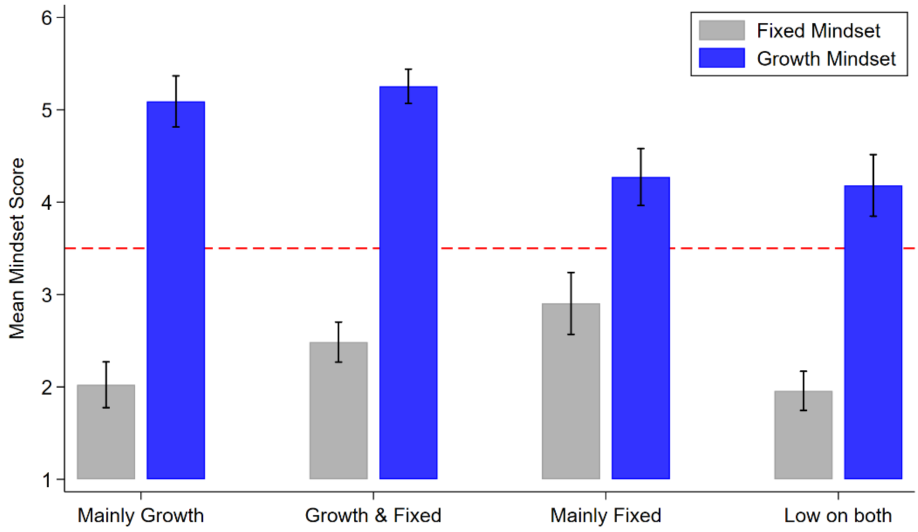
Participants in Group 1 ($N = 9$) demonstrated a mainly growth-oriented mindset, characterized by a high growth mindset score ($M = 5.09$, $SD = 0.28$) and a low fixed mindset score ($M = 2.02$, $SD = 0.25$). Group 2 ($N = 5$) included individuals with both high growth and high fixed mindset scores, with a mean growth mindset of 5.25 ($SD = 0.19$) and a mean fixed mindset of 2.49 ($SD = 0.22$). Group 3 ($N = 9$) consisted of participants with a mainly fixed mindset, as indicated by a lower growth mindset score ($M = 4.27$, $SD = 0.31$) and a higher fixed mindset score ($M = 2.90$, $SD = 0.34$). Finally, Group 4 ($N = 5$) included individuals who scored low on both the growth and fixed mindset scales, with a mean growth mindset of 4.18 (*SD* = 0.33) and a fixed mindset score of 1.96 (*SD* = 0.21).

Descriptive statistics showed that across all groups, fixed mindset scores were relatively low, with none of the groups averaging above the midpoint of the 6-point scale. A visual overview of the mindset group means is provided in Figure 8, clearly illustrating that, regardless of group classification, participants generally reported low fixed mindset beliefs and high growth mindset beliefs. This overall the results show that most participants tended to have a growth-oriented way of thinking.

Due to the limited variability, particularly the absence of consistently high fixed mindset scores, no further analyses were conducted on mindset as a moderator.

Figure 8

Mean Fixed and Growth Mindset Scores across The Four Mindset Groups



Note. The four mindset groups were created using a median split of participants' fixed and growth mindset scores separately: (1) Mainly growth (high growth, low fixed), (2) High growth and high fixed, (3) Mainly fixed (low growth, high fixed), and (4) Low on both. Error bars indicate standard deviations.

4 Discussion

Perceived effectiveness of feedback is crucial for supporting students' learning development, especially in reflective writing tasks. As AI is being used more in educational settings, it becomes important to understand how students respond to AI-generated feedback compared to human feedback, in order to ensure its most effective integration.

The present study aimed to investigate how the perceived feedback provider (human vs. AI) influenced students' trust and perceptions of feedback effectiveness in the context of written reflections. In order to test this, participants completed a reflective writing task and received the same feedback twice, once labeled as coming from a human and once as an AI chatbot. Furthermore, the study explored whether students' familiarity and frequency of interaction with AI chatbots influenced trust in the AI feedback provider. In addition, it explored the role of familiarity in the human assessor as well. Finally, the study examined the role of correctly or incorrectly identifying the perceived feedback provider.

4.1. Perceived feedback effectiveness and trust

This study's findings indicate that students generally perceived feedback as more effective when they believed it was provided by a human rather than generative AI. This aligns with prior literature showing a preference for human feedback over AI generated feedback (Hein et al., 2024; Şık and Pektaş, 2024; Steiss et al., 2024). The current results extend this literature by demonstrating that, even when the quality and content of feedback are held constant, perceptions of effectiveness are significantly influenced by the perceived identity of the feedback provider. This implies that the feedback content provided by generative AI may, in theory, be perceived as effective and possibly support students' learning development. However, the mere awareness that the feedback was generated by an AI chatbot seems to undermine this effect, reducing its impact and effectiveness of the feedback.

To gain insight into these observed effects, trust in the feedback provider was examined as a potential explanatory factor. The present findings show that students reported higher levels of trust in the human assessor than in the AI chatbot. This supports the finding of Bach et al. (2024) who found that in general people have greater trust in humans over AI systems. Importantly, all dimensions of trust (general trust, cognitive trust, and affective trust) were found to fully mediate the relationship between the perceived feedback provider and perceived effectiveness across multiple dimensions: perceived adequacy of the feedback, affective responses, and the feelings of empowerment. These findings support the idea that trust is a central component in feedback perception (Troy et al., 2024).

Despite the findings of Price et al. (2010), the present study showed that neither general nor cognitive trust mediated the effect of the perceived feedback provider on students' willingness to improve. Only affective trust demonstrated a partial mediation effect on this outcome. This suggests that the influence of trust indeed may differ depending on which aspect of trust is considered (Bach et al., 2022), and that motivational responses such as willingness to improve may be driven by other psychological or contextual factors beyond trust. For instance, prior research indicates that motivational constructs such as self-efficacy can play critical roles in determining whether individuals are willing to engage in improvement behaviors (Turda et al., 2021).

Interestingly, part of the current findings is in contrast with those of a recent study by Zhang et al. (2025), who found no significant differences in students' ratings of usefulness and objectivity for AI-generated feedback when the source was hidden versus revealed. These aspects, usefulness and objectivity, align with the construct of perceived adequacy of feedback. The current study, on the other hand, showed that revealing the feedback source led students to perceive AI-generated feedback as significantly less adequate than human feedback. These different results suggest that students' perceptions may vary depending on the context or the nature of the task (objective vs. subjective). Zhang et al. (2025) presented feedback from the different sources side by side, which may have led participants to base their evaluations more on the actual content of the feedback than the perceived source. Instead, this research controlled for direct content comparison, allowing for a clearer interpretation of how source perception alone influences feedback evaluation. Furthermore, differences in the measurements of objectivity and usefulness might account for variations between the study of Zhang et al. (2025) and the current study. Zhang et al. (2025) examined these two constructs separately with fewer items, while the present study included acceptance as a third construct to capture the overall perceived adequacy of the feedback. In addition, items such as the feeling of being satisfied and supported by the feedback were included to evaluate feedback usefulness and fairness more comprehensively. This study took a more holistic approach by calculating a mean score across multiple dimensions to reflect the overall adequacy of the feedback. As a result, it is possible that the additional items and the concept of acceptance played a significant role in predicting students' overall perceived adequacy of the feedback. Besides, Zhang et al. (2025) used a 5-point Likert scale to measure their constructs, whereas the current study used a 7-point Likert scale. Although both scales are widely used in research, the 7-point scale may allow for more detailed differences in students' feedback perceptions. Lastly, Zhang et al. (2025) focused on structured tasks with clear right or wrong answers, such as mathematics and programming, where feedback is more objective. The current study, however, focused on reflective writing. This is a more personal and subjective task (Mohamed et al., 2022; Ruijten-Dodoiu et al., 2024a), where trust might have a bigger impact (Mayer et al., 1995).

Overall, the present findings suggest that in contexts involving subjective reflection, trust in the feedback provider plays a key role in explaining the difference in how feedback effectiveness is perceived when it is labeled as coming from a human versus an AI chatbot. This underscores the importance of focusing on building trust in these AI systems to improve integrating AI in education effectively. The current study explored the relationship between students' trust and familiarity with AI chatbots as a potential approach.

4.2. Familiarity With the Feedback Provider

Specifically, the study explored whether familiarity and frequency of AI chatbot use influenced trust in the AI feedback provider. Contrary to some expectations (Luhmann, 1979; Gefen et al., 2003; Zhu et al., 2023), the present study did not find a significant positive relationship between source familiarity and trust. Specifically, the current findings show that AI chatbot familiarity, frequency of AI chatbot use, and frequency of AI chatbot use for feedback did not influence trust in the AI feedback provider. Similarly, no significant relationship was found between familiarity with the human assessor and trust in this person. However, the study did find a significant positive relationship between having the assessor as a teacher and both general and cognitive trust. This suggests that students who had the assessor as a teacher reported higher levels of general and cognitive trust than those who did not. While the study by Stöhr et al. (2024) found that familiarity plays a role in adoption and perceived usefulness, its effect on trust might be context dependent. This possible explanation is in line with recent work by Horowitz et al. (2024), who showed that while familiarity can enhance trust and acceptance of AI in some contexts, this effect is not universal and may depend on the specific application and ethical considerations. This suggests that, for subjective and personally relevant tasks like reflective writing, familiarity alone may be insufficient to increase trust in AI systems providing feedback.

4.3. Turing Test

Finally, the study examined whether participants could recognize the source of the feedback provider accurately through a Turing Test. Most participants (75%) correctly identified the feedback as AI-generated. Interestingly, individual differences in AI chatbot familiarity or frequency of use did not significantly predict the ability to correctly identify the source. This suggests that correctly recognizing AI feedback is not driven by experience alone, but could be influenced by other factors such as assumptions about the task and setting.

Further analysis examined whether accurate source identification moderated the relationship between the perceived feedback provider and students' evaluations of feedback effectiveness and trust. Results indicate that whether participants correctly or incorrectly identified the feedback source did not significantly moderate the effect of the condition on the different feedback effectiveness and trust scores. This suggests that even when participants correctly identified the source of the feedback, and may have been skeptical, there was still a significant change in their trust and perception of the feedback. This highlights how important the role of the initial belief of the feedback provider is.

4.4. Implications

The results of the present study have important implications for the integration of AI in educational feedback. If students' lower responses to AI-generated feedback compared to human feedback are not because of deficiencies in content quality, but rather from a lack of trust, then improving content alone may not be sufficient. Instead, educators and developers should focus on enhancing trust in AI feedback systems. Considering that trust in the feedback provider significantly influences multiple dimensions of feedback effectiveness, future research should investigate how technological and design features can support greater trust in AI generated feedback. Such insights are essential for ensuring AI feedback is not only detailed and consistent but also perceived as effective and supportive to students' learning development, especially in reflective writing contexts.

To better understand why students perceive AI feedback as less effective in terms of willingness to improve, future research should explore additional motivational and psychological factors beyond trust. Constructs such as self-efficacy, and emotional reactions may influence feedback engagement and improvement behaviors (Turda et al., 2021). Investigating these variables could help identify the mechanisms underlying students' motivation to improve and inform which aspects of AI feedback should be prioritized to effectively support students' learning outcomes.

Even though AI familiarity did not have a direct impact on trust in the AI feedback provider in this study, it may play a more nuanced moderating role. What if AI familiarity might influence the extent to which trust in the source affects the perceived effectiveness of feedback. Specifically, AI familiarity could still play a moderating role by reducing the negative effect of students' lower trust in AI chatbots on their perceptions of feedback effectiveness. Future research should explore whether students with greater AI familiarity show a smaller difference in the effect of trust on feedback perceptions between human and AI sources. Investigating this potential moderating effect could provide insights into whether increasing AI familiarity is a useful strategy for enhancing students' perceptions of AI-generated feedback effectiveness.

Furthermore, other factors could have explained students' lower trust ratings in the AI feedback provider. Specifically, there are known to be biases against AI such as the belief that it lacks emotional intelligence, perceived competence, or transparency (Hutson & Plate, 2024; Nguyen, 2023; Maity & Deroy, 2024). Kahr et al. (2025) and Shahzad et al. (2024) emphasize the importance of transparency and explainability in building trust. Supporting this, Ruijten-Dodoiu et al. (2025) state that a lack of transparency can lead to distrust or misconceptions about the reliability of AI-generated feedback, potentially undermining its effectiveness. Furthermore, prior research suggests that emotional responsiveness, empathy, and psychological support offered by human teachers remain highly valued by students (Nguyen, 2023; Rüdian et al., 2025). Given the subjective nature of reflective tasks, these interpersonal components might be particularly important in this context. Therefore, further research should focus on these targets for improving trust in AI feedback providers, and how this affects feedback effectiveness perceptions.

Considering all these points, for effective integration of AI in education, it is necessary to not only ensure high-quality feedback content, but also to build and maintain students' trust in AI systems as feedback providers. Increasing trust through enhancing transparency, explainability, and interpersonal components, might improve students' perceptions of AI feedback effectiveness and, in turn, support their learning outcomes.

4.5. Limitations and Future Research

Several limitations of the present study should be acknowledged.

Although the sample size was sufficient to detect large effects, it may have lacked power to detect small or moderate effects even if they exist. Especially for the predictors AI familiarity and frequency of use on the trust variables. As a result, the non-significant findings should be interpreted with caution, as they may reflect insufficient power rather than a true absence of relationships between the predictors and general trust in AI. Additionally, the sample was primarily drawn from the experimenters' social circles and consisted of only people studying at Eindhoven University of Technology. This limits the generalizability of findings beyond technical or Dutch university settings. However, the use of a well-defined academic task, reflective writing, enhances ecological validity. Future research should include larger and more diverse samples to enhance generalizability and assess the roles of AI familiarity and frequency of use more reliably.

Furthermore, as conducting multiple mediation analyses allowed for the exploration of unique comparisons, the findings should be interpreted with caution. Given the large number of analyses conducted, combined with high correlations between some variables, this could have increased the likelihood of finding significant effects by chance.

The analyses that were used in this study were suited for the research question, but they mostly rely on the assumption of linear relationships between predictors and outcomes. This might not fully reflect the complex nature of learning and perception. Therefore, future research might consider using nonlinear models or other more suitable approaches to provide a more comprehensive understanding of AI feedback in educational contexts.

In addition, the initial plan included exploring whether students' mindset (fixed vs. growth) moderated the effect of perceived feedback provider on perceived feedback effectiveness, however this analysis was not conducted. This was due to limited variability in mindset scores, particularly the lack of consistently high fixed mindset scores, possibly because many university students tend to have a growth mindset (Jeffs et al., 2021; Prescott et al., 2024). However, mindset remains a potentially important factor influencing how students perceive feedback effectiveness. Future research should further investigate the role of mindset, especially in relation to human versus AI-generated feedback, to better understand its impact on motivation and acceptance among students.

Even though analyses show that participants who correctly identified the feedback source did not differ significantly in trust levels between the feedback providers compared to those who guessed incorrectly, the overall reduced trust in AI-generated feedback might still be influenced by the deception-based design. This design initially led participants to believe the feedback was human-generated before revealing it was AI-generated. Such deception may have affected participants' responses through feelings of mistrust, which could have potentially influenced their trust ratings. Further research should include a between-participants design controlling for order effects.

Moreover, current AI systems can struggle with multiple criteria instructions. Therefore, variations in prompt engineering, such as the multiple rubrics and student answers, which may have caused differences in prompt length, may have introduced inconsistencies in AI feedback quality and its output. Although the within-subjects design helped reduce these concerns, future studies should aim for even more consistent feedback prompts.

This study focused on trust rather than credibility because trust might play a more immediate role than perceived expertise in the context of reflective writing. However, future research could integrate both dimensions to assess their independent and combined influence on feedback perception. Understanding how trust and credibility interact can provide deeper insights into how students perceive feedback, and help identify the key factors that contribute to their perceptions of feedback effectiveness.

Finally, future research could build on the proposal of Ruijten-Dodoiu et al. (2025) to extract longitudinal insights across a longer period of time instead of one reflective writing task. This could provide deeper understanding of how AI generated feedback influences student development beyond isolated tasks. Additionally, this approach might help create more personalized learning paths, tailored support, and encourage ongoing reflective growth.

5 Conclusion

In summary, the present study demonstrated that students perceive feedback as more effective when it is believed to be provided by a human versus AI, even when feedback quality is held constant. Trust in the feedback provider emerged as a key mediator of this relationship across multiple dimensions of feedback effectiveness, including perceived adequacy, affective response, and feelings of empowerment. These findings extend prior research by emphasizing the critical role of trust in shaping feedback perception, particularly in subjective, reflective writing contexts. These results highlight the important role of building trust in AI systems to enhance their effective integration in education. This study's findings showed that students' familiarity with AI chatbots or their frequency of use did not significantly influence trust, indicating that simply increasing exposure to AI chatbots may not be enough to build trust in educational settings.

Future research should investigate how trust in AI feedback can be enhanced through transparency, explainability, and interpersonal components, to ultimately increase students' perceptions of feedback effectiveness. This is important given the crucial role that effective feedback plays in supporting students' reflective processes and overall learning development. Additionally, research should explore how AI-generated feedback specifically affects students' psychological and motivational factors, such as self-efficacy, which influence their willingness to improve after receiving feedback. Understanding these effects is important for designing AI feedback systems that not only build trust but also effectively motivate students to improve. Furthermore, larger and more diverse samples might be necessary for future research, especially to assess the influence of AI chatbot familiarity and frequency of use, and individuals' mindset (fixed vs. growth) more reliably. Lastly, longitudinal studies could examine how feedback perceptions evolve over time and across different types of learning tasks, paving the way for more effective integration of AI-generated feedback that supports students' reflective learning and development.

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Use of AI tools In this study, I used ChatGPT to assist me with rewriting some text to be more academic and coherent. This was especially helpful for rephrasing my own reasoning and conclusions, which were often written down in a mixture of English and Dutch. ChatGPT helped me translate and formulate these into correct English. I still wrote it down myself in a way that I felt was most appropriate. Additionally, ChatGPT provided valuable insights and answers related to the analyses used in the results section. Finally, I used ChatGPT to format the letters associated with the results in the results section in italics.

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Appendix A Assessment Rubric

This rubric was adapted from Erasmus University Rotterdam (2023) and serves as a guideline for assessing reflective writing in accordance with established academic standards.

Criteria

1. Description
   - Excellent (4): Clear and concise description of the topic with comprehensive details and an objective overview.
   - Good (3): Adequate description of the topic with relevant details.
   - Fair (2): Somewhat vague description with limited details.
   - Poor (1): Lacks clarity and coherence in describing the topic.
2. Feelings
   - Excellent (4): Reflects deep understanding and insightful reflection on personal feelings, thoughts, and experiences related to the topic.
   - Good (3): Expresses reasonable understanding and provides some reflection on personal feelings and thoughts related to the topic.
   - Fair (2): Shares limited feelings and thoughts with some gaps in reflection.
   - Poor (1): Lacks reflection or feeling related to the topic.
3. Evaluation
   - Excellent (4): Offers comprehensive and insightful evaluation of the subject matter with critical analysis and supported judgments.
   - Good (3): Provides a fair evaluation of the subject matter with some analysis and reasonable judgments.
   - Fair (2): Offers limited evaluation of the subject matter with gaps in analysis and judgments.
   - Poor (1): Does not evaluate the subject matter or provides inaccurate evaluation.
4. Analysis
   - Excellent (4): Analyzes the topic thoroughly, identifying and discussing key factors, causes, and impacts, supported by clear examples and evidence.
   - Good (3): Provides a reasonable analysis of the topic, identifying and discussing important factors, causes, and impacts with supporting examples.
   - Fair (2): Offers limited analysis of the topic, with some gaps in identifying and discussing factors, causes, and impacts.
   - Poor (1): Does not provide analysis or provides inaccurate analysis of the topic.
5. Conclusion
   - Excellent (4): Presents a clear and concise conclusion that effectively summarizes the key takeaways or lessons learned from the reflection.
   - Good (3): Offers a reasonable conclusion that summarizes the main takeaways or lessons learned from the reflection.
   - Fair (2): Provides a somewhat vague or incomplete conclusion that lacks clarity or fails to effectively summarize the key takeaways.
   - Poor (1): Lacks conclusion or provides a conclusion that does not effectively summarize the key takeaways.
6. Next Steps
   - Excellent (4): Provides specific and actionable next steps for improvement or future actions based on the reflection, demonstrating clear insight and logical progression.
   - Good (3): Suggests reasonable next steps for improvement or future actions based on the reflection, with some clarity and logical progression.
   - Fair (2): Offers general or vague next steps for improvement or future actions, lacking detail or specificity.
   - Poor (1): Lacks clarity or specificity in suggesting next steps for improvement or future actions.

Appendix B Measures

Feedback perception questionnaire

Items of the Feedback Perceptions Questionnaire (FPQ) adapted from Strijbos et al. (2010), answered on a 7-point response format (1 = fully disagree, 7 = fully agree).

Perceived Adequacy of Feedback (PAF)
"I would be satisfied with this feedback"
"I would consider this feedback fair"
"I would consider this feedback justified"
"I would consider this feedback useful"
"I would consider this feedback helpful"
"This feedback would provide me a lot of support"
"I would accept this feedback"
"I would accept this feedback"
"I would dispute (/disagree with) this feedback"
"I would reject this feedback"

Willingness to Improve (W)
"I would be willing to improve my performance"
"I would be willing to invest a lot of effort in my revision"
"I would be willing to work on further reflective writing assignments"

Affect (A)
"I would feel satisfied receiving this feedback on my reflective task."
"I would feel confident receiving this feedback on my reflective task."
"I would feel successful receiving this feedback on my reflective task."
"I would feel offended receiving this feedback on my reflective task."
"I would feel angry receiving this feedback on my reflective task."
"I would feel frustrated receiving this feedback on my reflective task."

Feedback effectiveness questionnaire

Items of the Feedback Effectiveness Scale (FES) adapted from Lizzio and Wilson (2008), answered on a 7-point response format (1 = not at all, 7 = very).

"Overall, how effective has the feedback, provided by the assessor, been in facilitating your learning?"
"Overall, how effective has the feedback, provided by the assessor, been in facilitating your competence as a learner?"
"Overall, how effective has the feedback, provided by the assessor, been in facilitating your confidence as a learner?"

Trust questionnaire

One single-item questionnaire to measure general trust adapted from Yin et al. (2019), answered on a 5-point response format (1 = no trust at all, 5 = full trust).

General Trust (GT)
"How much do you trust this assessor to provide you with the guidance and service you need?"

Items of the Semantic Differential Scale (SDS) adapted from Shang et al. (2024), answered on a 5-point response format.

Cognitive Trust (CT)
"Unreliable - Reliable"
"Inconsistent - Consistent"
"Unpredictable - Predictable"
"Undependable - Dependable"
"Careless - Careful"
"Clueless - Knowledgeable"
"Incompetent - Competent"
"Ineffective - Effective"
"Amateur - Proficient"
"Irrational - Rational"
"Unreasonable - Reasonable"
"Incomprehensible - Understandable"
"Dishonest - Honest"
"Unfair - Fair"

Affective Trust (AT)
"Apathetic - Empathetic"
"Insensitive - Sensitive"
"Impersonal - Personal"
"Ignoring - Caring"
"Rude - Cordial"
"Indifferent - Responsive"

Familiarity AI

Items of frequency of use of chat-based AI and familiarity adapted from Elhassan et al. (2025).

How familiar are you with chat-based AI softwares?
Not familiar at all – a little – moderately – very familiar - extremely familiar

How frequently do you use chat-based AI softwares for educational purposes?
Never – rarely – monthly – weekly - daily

How frequently do you ask chat-based AI softwares to give you feedback on your written work?
Never – rarely – monthly – weekly - daily

Familiarity human

How well do you know the assessor?
I don't know him/her – a little – moderately – very well - extremely well

Did you have this assessor as a teacher in the previous quartile (Q3) or in the current quartile (Q4)?
Neither Q3 nor Q4, Only Q3, Only Q4, both Q3 and Q4

Mindset questionnaire

Items of the Multidimensional Mindset Scale (MUMIS) adapted from Alvarado et al. (2024), answered on a 6-point response format (1 = strongly disagree, 6 = strongly agree).

“I can be more intelligent if I work on it”
“My level of intelligence cannot be modified because it always has been the same”
“My level of intelligence has changed over time”
“I am an intelligent person because I was born that way and I cannot modify it”
“I believe I could prepare myself to solve logical and numeric problems to advance my math level”
“I believe I could prepare myself to play an instrument that I have never played”
“I believe I could prepare myself to participate in any kind of sport challenge”
“I believe I could prepare myself to start conversations with strangers and develop social relationships with them”
“I believe I could identify and recognize my own feelings, needs and motivations to comprehend myself”
“I believe writing an essay or communicating orally are and will not be my strengths”
“I believe I cannot do much about my sense of rhythm and music skills”
“I believe I am not skilled to exercise, and I cannot do much about it”
“I believe I am not good enough to interact with others, and I will not be able to modify this”
“I believe I am not good enough to recognize my own's desires and feelings, and I cannot modify this”
“I prefer to avoid challenges”
“Intelligent people do not need to make an effort to succeed”
“I do not really care about learning, while obtaining the results I want”
“I do not think practicing helps intelligent people to improve”
“I feel attracted to challenges”
“I believe in effort to become better”
“Practicing is necessary to improve any results”
“I am willing to participate whenever there is a challenge”
“I tend to give up when facing a difficult task or situation”
“I tend to experience negative emotions and feelings when I am confronted with difficulty”
“I feel incapable of changing my abilities when I feel frustrated”

Appendix C: Informed Consent Form

EINDHOVEN UNIVERSITY OF TECHNOLOGY

Information sheet for research project "Perceptions of feedback on written reflections"

1. Introduction

You have been invited to take part in research project "Perceptions of feedback on written reflections", because you are registered in the participant database.

Participation in this research project is voluntary: the decision to take part is up to you. Before you decide to participate we would like to ask you to read the following information, so that you know what the research project is about, what we expect from you and how we deal with processing your data. Based on this information you can indicate via the consent declaration whether you consent to take part in this research project and the processing of your data.

2. Purpose of the research

This research project will be managed by Petra Slokker, a student under supervision of dr.ir Peter Ruijten-Dodoiu. The purpose of this research project is to explore how students perceive and evaluate feedback from human experts on their written reflections.

3. Controller in the sense of the GDPR

TU/e is responsible for processing your personal data within the scope of the research. The contact details of TU/e are: Technische Universiteit Eindhoven, De Groene Loper 3, 5612 AE Eindhoven.

4. What will taking part in the research project involve?

You are asked to complete a reflective writing task and to fill in several questionnaires. You will be asked several questions about your perceptions of that feedback and its source. The duration of the study will be at most 45 minutes.

This study will be completely anonymous, and the data obtained from the study will not be traceable to you nor will be seen by the assessor. Anything you will write in the reflective task will also be sent to the assessor in anonymized way; they will never know it was you who wrote that text.

For your participation in this research project you will receive a compensation of €9 via Tikkie payment. Your personal information that is related to this will be removed after payment is completed.

5. Potential risks and inconveniences

Your participation in this research project does not involve any physical, legal or economic risks. You do not have to answer questions which you do not wish to answer. Your participation is voluntary.

6. Confidentiality, use, and storage of data

We will do everything we can to protect your privacy as best as possible. Within the framework of the research project we do not process any personal data. The research results that will be published will not in any way contain confidential information or personal data from or about you through which anyone can recognize you.

The raw and processed research data will be retained for a period of 10 years. The research data will, if necessary (e.g. for a check on scientific integrity) and only in anonymous form be made available to persons outside the research group.

7. Withdrawing your consent and contact details

Participation in this research project is entirely voluntary. You may end your participation in the research project at any moment, or withdraw your consent to using your data for the research, without specifying any reason. Ending your participation will have no disadvantageous consequences for you.

Do you wish to end the research, or do you have any questions and/or complaints? Then please contact the researcher during this study, or for later contact via p.slokker@student.tue.nl.

8. Legal ground for processing your personal data

The legal basis upon which we process your data is consent.

This research project was assessed and approved by the ethical review committee of the HTI group of the Eindhoven University of Technology.

NOTE: The consent form was presented at the start of the experiment; participants could indicate their consent by signing their name.

Consent form for participation

By signing this consent form I acknowledge the following:

1. I am sufficiently informed about the research project through a separate information sheet. I have read the information sheet and have had the opportunity to ask questions. These questions have been answered satisfactorily.

2. I take part in this research project voluntarily. There is no explicit or implicit pressure for me to take part in this research project. It is clear to me that I can end participation in this research project at any moment, without giving any reason. I do not have to answer a question if I do not wish to do so.

Certificate of consent

I, (NAME) ……
consent to participate in this study.

Participant's Signature

Date